LEMMA ALGEBRAIZATION PRINCIPAL BIS

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter algebraization.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \external-document list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-lemma-algebraization-principal-bis.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original reference 0EHC: sources/algebraization.tex, source lines 4983–4983. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 087X: sources/coherent.tex, source lines 6313–6313. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 0EIU: sources/algebraization.tex, source lines 5458–5458. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 0EH9: sources/algebraization.tex, source lines 220–220. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EHH: sources/algebraization.tex, source lines 5328–5328. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EHA: sources/cohomology.tex, source lines 9262–9262. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0DXT: sources/algebraization.tex, source lines 4997–4997. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EHK: sources/algebraization.tex, source lines 6381–6381. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 031D: sources/algebra.tex, source lines 22860–22860. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0GYK: sources/cohomology.tex, source lines 8931–8931. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *In Situation 0EHC let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Assume*

(1) *A is local and $\mathfrak{a} = \mathfrak{m}$ is the maximal ideal,*
(2) *A has a dualizing complex,*
(3) *$I = (f)$ is a principal ideal,*
(4) *$(\mathcal{F}_n)$ satisfies the $(2, 3)$-inequalities.*

Then $(\mathcal{F}_n)$ extends to X. In particular, if A is I-adically complete, then $(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_U$-module.

Proof. Recall that $Coh(U, I\mathcal{O}_U)$ is an abelian category, see Cohomology of Schemes, Lemma 087X. Over affine opens of U the object $(\mathcal{F}_n)$ corresponds to a finite module over a Noetherian ring (Cohomology of Schemes, Lemma 087W). Thus the kernels of the maps $f^N : (\mathcal{F}_n) \to (\mathcal{F}_n)$ stabilize for N large enough. By Lemmas 0EIU and 0EIL in order to prove the lemma we may replace $(\mathcal{F}_n)$ by the image of such a map. Thus we may assume f is injective on $(\mathcal{F}_n)$. After this replacement the equivalent conditions of Lemma 0EH9 hold for the inverse system $(\mathcal{F}_n)$ on U. We will use this without further mention in the rest of the proof.

We will check hypotheses (a), (b), and (c) of Lemma 0EHH. Hypothesis (b) holds by Cohomology, Lemma 0EHA.

Pick an inverse system of modules $\{M_n\}$ as in Lemma 0DXT. We may assume $H^0_{\mathfrak{m}}(M_n) = 0$ by replacing M_n by $M_n/H^0_{\mathfrak{m}}(M_n)$ if necessary. Then we obtain short exact sequences

$$0 \to M_n \to H^0(U, \mathcal{F}_n) \to H^1_{\mathfrak{m}}(M_n) \to 0$$

for all n. Let E be an injective hull of the residue field of A. By Lemma 0EHK and our current assumption (4) we can choose, an integer $m \geq 0$, finite A-modules N_1 and N_2 annihilated by f^c for some $c \geq 0$ and compatible systems of maps

$$H^i_{\mathfrak{m}}(M_n) \to \mathrm{Hom}_A(N_i, E), \quad i = 1, 2$$

for $n \geq m$ with the properties stated in the lemma.

We know that $M = \lim H^0(U, \mathcal{F}_n)$ is an A-module whose limit topology is the f-adic topology. Thus, given n, the module M/f^nM is a subquotient of $H^0(U, \mathcal{F}_N)$ for some $N \gg n$. Looking at the information obtained above we see that f^cM/f^nM is a finite A-module. Since f is a nonzerodivisor on M we conclude that $M/f^{n-c}M$ is a finite A-module. In this way we see that hypothesis (c) of Lemma 0EHH holds.

Next, we study the module

$$Ob = \lim H^1(U, \mathcal{F}_n) = \lim H^2_{\mathfrak{m}}(M_n)$$

For $n \geq m$ let K_n be the kernel of the map $H^2_{\mathfrak{m}}(M_n) \to \mathrm{Hom}_A(N_2, E)$. Set $K = \lim K_n$. We obtain an exact sequence

$$0 \to K \to Ob \to \mathrm{Hom}_A(N_2, E)$$

By the above the limit topology on $Ob = \lim H^2_{\mathfrak{m}}(M_n)$ is the f-adic topology. Since N_2 is annihilated by f^c we conclude the same is true for the limit topology on $K = \lim K_n$. Thus K/fK is a subquotient of K_n for $n \gg 1$. However, since $\{K_n\}$ is pro-isomorphic to an inverse system of finite length A-modules (by the conclusion of Lemma 0EHK) we conclude that K/fK is a subquotient of a finite length A-module. It follows that K is a finite A-module, see Algebra, Lemma 031D. (In fact, we even see that $\dim(\mathrm{Supp}(K)) = 1$ but we will not need this.)

Given $n \geq 1$ consider the boundary map

$$\delta_n : H^0(U, \mathcal{F}_n) \longrightarrow \lim_N H^1(U, f^n\mathcal{F}_N) \xrightarrow{f^{-n}} Ob$$

(the second map is an isomorphism) coming from the short exact sequences

$$0 \to f^n\mathcal{F}_N \to \mathcal{F}_N \to \mathcal{F}_n \to 0$$

For each n set

$$P_n = \mathrm{Im}(H^0(U, \mathcal{F}_{n+m}) \to H^0(U, \mathcal{F}_n))$$

where m is as above. Observe that $\{P_n\}$ is an inverse system and that the map $f : \mathcal{F}_n \to \mathcal{F}_{n+1}$ on global sections maps P_n into P_{n+1}. If $p \in P_n$, then $\delta_n(p) \in K \subset Ob$

because $\delta_n(p)$ maps to zero in $H^1(U, f^n\mathcal{F}_{n+m}) = H^2_{\mathfrak{m}}(M_m)$ and the composition of δ_n and $Ob \to \mathrm{Hom}_A(N_2, E)$ factors through $H^2_{\mathfrak{m}}(M_m)$ by our choice of m. Hence

$$\bigoplus\nolimits_{n\geq 0} \mathrm{Im}(P_n \to Ob)$$

is a finite graded $A[T]$-module where T acts via multiplication by f. Namely, it is a graded submodule of $K[T]$ and K is finite over A. Arguing as in the proof of Cohomology, Lemma 0GYK[1] we find that the inverse system $\{P_n\}$ satisfies ML. Since $\{P_n\}$ is pro-isomorphic to $\{H^0(U, \mathcal{F}_n)\}$ we conclude that $\{H^0(U, \mathcal{F}_n)\}$ has ML. Thus hypothesis (a) of Lemma 0EHH holds and the proof is complete. □

[1]Choose homogeneous generators of the form $\delta_{n_j}(p_j)$ for the displayed module. Then if $k = \max(n_j)$ we find that for $n \geq k$ and any $p \in P_n$ we can find $a_j \in A$ such that $p - \sum a_j f^{n-n_j} p_j$ is in the kernel of δ_n and hence in the image of $P_{n'}$ for all $n' \geq n$. Thus $\mathrm{Im}(P_n \to P_{n-k}) = \mathrm{Im}(P_{n'} \to P_{n-k})$ for all $n' \geq n$.